Drift to Default: Long Autoregressive Video Rollouts Relax to a Prompt-Set State

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Abstract

Autoregressive video diffusion models generate long videos chunk by chunk, conditioned on their own output, and drift over tens of seconds. We show that this drift is not a random walk but *relaxation*. The colour and texture statistics of a rollout move from those of its opening toward a default set by the text prompt, and the long-run state is an equilibrium between two pulls:

$$x_\infty \approx \alpha\, x_{\text{open}} + (1-\alpha)\, \mu_{\text{prompt}}.$$

- **Generality.** About 22–40% of an opening's deviation survives 31 s. This holds for brightness, colour, saturation, contrast and sharpness, across [9] model configurations spanning DMD, SiD, GAN, LongLive, CausVid, Rolling Forcing and SkyReels diffusion forcing. The default μ_{prompt} moves with the lighting described in the prompt (16–45 L^*) and differs between models for the same words.
- **Mechanism.** An attention sink keeps the opening in the context and sets α. Fitted to the observed persistence curves, a linear chunk-recursion model recovers the sink's per-chunk weight that we measure directly by bit-exact KV-cache replay (0.05). The sink's tiny direct weight still sets the equilibrium because the prompt's per-chunk pull is equally small.
- **Two levers.** Raising α with a training-free, content-aware correction reduces drift and keeps 83–93% of legitimate brightness changes; post-hoc colour matching keeps 7–12%. Moving μ_{prompt} toward the opening with a lighting description calibrated to measured brightness reduces drift by 26% on Self Forcing; a free-form VLM caption does not.

All experiments use paired seeds on a pinned GPU architecture. Six analyses were preregistered with hashed code, and we report the two that failed.

1 Introduction

Causal (autoregressive) video diffusion models [1, 2] make streaming, open-ended generation possible and underlie interactive world models. Their known weakness is drift: over tens of seconds the picture darkens, saturates, loses texture or collapses. Many recent methods target this failure, through rolling windows, attention sinks, re-aligned positions, longer self-rollouts and test-time correction [7, 8, 9, 12, 13]. They are usually evaluated with one sample per prompt, an aggregate quality score and no uncertainty. What drift actually *is* has not been measured.

We treat drift as a measurement problem and borrow the design tools of clinical research: paired seeds, controlled interventions, held-out prompts, cluster-bootstrap intervals and preregistered analyses. Our central finding is simple. A long rollout *relaxes*: every low-level statistic moves from its opening value toward a default set by the prompt, and a sink holds back a fixed share of the opening. This one picture explains several observations that otherwise look unrelated:

- why sinks reduce drift;
- why real camera frames "drift more" than generated ones;
- why a test-time correction that copies the anchor's tone works but over-corrects at early denoising steps;
- why describing the opening in the prompt fails while calibrating the lighting words succeeds.

Contributions.

1. A measurement protocol for long-video drift: self-referenced metrics, held and *perturbed* openings, paired seeds pinned to one GPU architecture, and preregistration. It also exposes three evaluation pitfalls:
   - the GPU architecture changes bf16 outputs;
   - resumed jobs silently break pairing;
   - a standard image-quality metric is blind to drift.
2. The relaxation law, $x_\infty \approx \alpha x_{\text{open}} + (1-\alpha)\mu_{\text{prompt}}$. We establish it interventionally, by shifting the brightness and colour of the same opening, across [9] model configurations and six statistics.
3. A quantitative mechanism. Long-horizon persistence curves and direct KV-cache interventions give consistent per-chunk weights, which explains how a sink with a tiny weight sets the equilibrium.
4. Two practical levers matching the two terms of the law, both evaluated in preregistered tests against baselines: a content-aware correction raises α, and calibrated prompts move μ.

No model is trained. Everything is inference on public weights, so the protocol applies to any open autoregressive video model.

2 Related work

Autoregressive video diffusion.

- Diffusion Forcing [3] trains with independent per-frame noise.
- CausVid [1] distils a bidirectional teacher [4] into a causal few-step student with DMD [5].
- Self Forcing [2] trains on the model's own rollouts.
- Long-video successors add rolling KV windows, attention sinks (as in streaming language models [11]), RoPE re-alignment, LoRA-based extension, and joint rolling denoising [7, 8, 9].
- SkyReels-V2 [10] extends videos by diffusion forcing over overlapping segments.

Error accumulation and correction. Meta-ARVDM [14] analyses error accumulation theoretically. Pathwise test-time correction [12] and FreqForcing [13] correct drift at inference; our correction is a deliberately minimal member of this family. It transfers only the DC tone at one early step, which lets us study the step as a knob.

Evaluation. VBench [15] scores imaging quality with MUSIQ [16]. We show that MUSIQ is nearly blind to drift, and we measure drift relative to each prompt's own fresh-start distribution, in colour statistics and in DINOv2 [17] features.

3 Setup

Models. All configurations generate 832×480 video at 16 fps in chunks of 3 latent frames (0.75 s), except SkyReels (960×544, 24 fps); details are in Table 1.

- **Self Forcing [2] checkpoints:** DMD (main), SiD, SiD-v2, GAN, and the 10-second variant. All share one architecture (Wan2.1-1.3B [4]) and differ only in the distillation objective.
- **CausVid [1]:** autoregressive DMD checkpoint.
- **LongLive [7]:** LoRA, official window 12 + sink 3.

Table 1: Model configurations. All except SkyReels share the Wan2.1-T2V-1.3B architecture and are run in our harness (`hold_gen.py`, bit-identical to the official Self Forcing loop when no perturbation is applied); window and sink sizes are in latent frames (3 per chunk of 0.75 s).

Configuration	Training / generation	Steps	Context
Self Forcing DMD	self-rollout, DMD [5]	4	window 21; sink 0 / 1 / 3
Self Forcing SiD, SiD v2	self-rollout, SiD [6]	4	window 21, sink 3
Self Forcing GAN	self-rollout, GAN	4	window 21, sink 3
Self Forcing 10 s	self-rollout on 10-s clips	4	window 21, sink 3
CausVid [1]	teacher-forced DMD	3	window 21, sink 3
LongLive [7]	long-rollout LoRA	4	window 12; sink 0 / 3
Rolling Forcing [8]	rolling joint denoising	5	anchor block + 21
SkyReels-V2-DF [10]	diffusion forcing, 544p/24 fps	30 + CFG	overlap 17 frames

- **Rolling Forcing [8]:** rolling joint denoising with a fixed anchor block.
- **SkyReels-V2-DF-1.3B [10]:** diffusion forcing over overlapping segments.

Held and perturbed openings. To separate drift from what the prompt asks for, an opening image is held for 0.56 s, encoded as the first chunk (or as the prefix video for SkyReels), and continued for 31 s with a generic driving prompt. Openings are NAVSIM [19] front-camera frames or first frames generated by the model. In the interventional study, the same opening is shifted in CIELAB before encoding: L^* by ± 8 and ± 16, and b^* by ± 12.

Metrics. For each video we record per-frame Lab means, saturation, L^* contrast and Laplacian sharpness. The *persistence* α is the within-scene slope of the end value (last 2 s) on the opening value. For prompt-only rollouts, drift is measured against 10 fresh 5-second generations of the same prompt: D is the Mahalanobis distance of colour statistics and K the DINOv2 nearest-neighbour distance.

Statistics. Seeds are paired across arms, and every arm of a scene runs in one job on one GPU architecture. We use mixed models or seed-averaged paired differences, with scene-cluster bootstrap intervals and Holm correction within each preregistered family. Each preregistration froze its code by SHA-256 hash before generation, and the last four were committed publicly first.

4 Drift is relaxation toward a prompt-set default

Observational evidence. Across 40 held openings with a generic prompt, the change of every statistic over 31 s is strongly anti-correlated with its opening value (Table 2): r between -0.79 and -0.95 for L^*, a^*, b^*, saturation, sharpness and contrast, on both Self Forcing and LongLive. For brightness:

$$\Delta L = 22.8 - 0.69\, L_{\text{open}}, \quad R^2 = 0.82.$$

- Only 22–40% of an opening's deviation survives.
- Real and generated openings fall on the same line ($p = 0.80$), so real frames "drift more" only because they are brighter.

The default is set by the prompt. Continuing the same openings under four lighting prompts gives four levels (Fig. 1b). On Self Forcing these are sunny 43, overcast 34, dusk 19 and night 16 L^*; the lighting term has $p \approx 5 \times 10^{-78}$. On LongLive the same words give 45, 40, 32 and 21. The prompt sets the target, and the mapping from words to brightness is model-specific.

Table 2: Every low-level statistic relaxes toward a default (40 held openings, generic prompt). Persistence = 1 + slope of the 31-s change on the opening value; r = correlation of change with opening value.

Statistic	Self Forcing persistence	r	LongLive persistence	r
L^*	0.31	-0.91	0.26	-0.81
a^*	0.38	-0.92	0.39	-0.90
b^*	0.22	-0.88	0.42	-0.80
saturation	0.54	-0.92	0.23	-0.95
log sharpness	0.37	-0.91	0.23	-0.94
L^* contrast	0.34	-0.82	0.37	-0.79

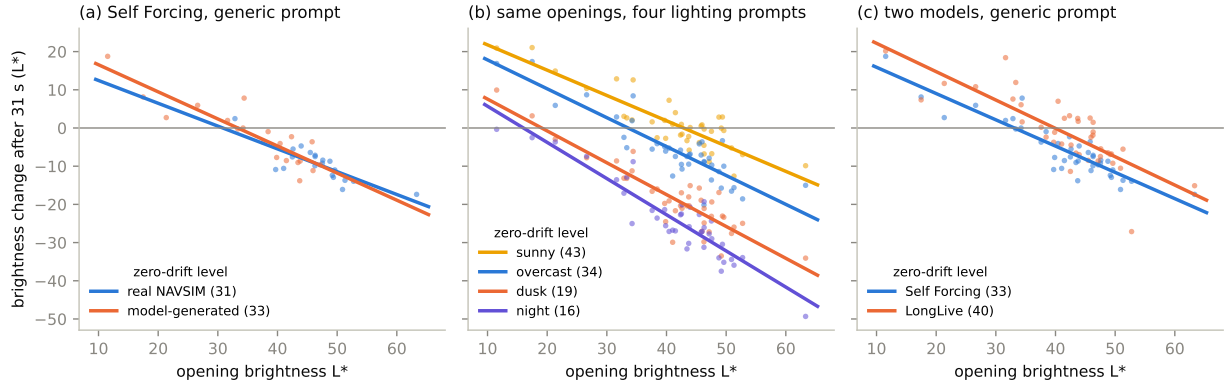


Figure 1: Brightness change after 31 s against the opening's brightness (one dot per opening). (a) Real and generated openings follow one line. (b) Four lighting prompts set four defaults. (c) Two models: same slope, different default.

Interventional evidence across models. [Relaxation study (preregistered): 24 scenes × 7 opening versions × 9 configurations. Within-scene α_L, α_b per configuration; H1–H4; table of α, half-life and default level per model; fresh-start comparison; long-horizon subset.]

5 Mechanism: an equilibrium between the anchor and the prompt

Model a chunk's mean statistic as a linear recursion:

$$x_{k+1} - \mu = w_w\big(\bar{x}_{\text{window}} - \mu\big) + w_s\big(x_{\text{sink}} - \mu\big).$$

The opening counts as a window chunk until it rolls out; after that, with a sink, it keeps a direct pull w_s. The prompt's pull is $w_p = 1 - w_w - w_s$. Without a sink the opening is eventually forgotten ($\alpha \to 0$). With a sink the equilibrium share is

$$\alpha_\infty = \frac{w_s}{w_s + w_p}.$$

The share is large whenever the prompt's pull is as weak as the sink's.

Two independent measurements agree. Bit-exact KV-cache replay lets us shift only the sink or only the window and regenerate the next chunk from identical noise. It measures $w_s = 0.053$ $[0.040, 0.066]$ for synthetic tone shifts, 0.036 in a dose-response design, and $w_w \approx 0.76$ (synthetic) to 0.90–0.99 (natural differences). Fitting the recursion to the long-horizon persistence curve alone gives:

- Self Forcing: $w_s = 0.049$ $[0.005, 0.069]$, $w_w = 0.85$, $\alpha_\infty = 0.32$;
- LongLive: $w_s = 0.054$, $w_w = 0.81$.

Table 3: Correction versus post-hoc colour matching on the preregistered confirmation set (40 held-out prompts × 4 seeds; transitions: 8 held-out prompts with legitimate brightness changes).

Method	ΔD [95% CI]	Δz_L	transitions kept	jitter
Content-aware correction, $\beta = 1$	-1.48 [-1.76, -1.19]	+1.08	0.83	×1.13
Content-aware correction, $\beta = 1.5$	-1.78 [-2.11, -1.45]	+1.32	0.93	×1.67
Post-hoc mean matching	-0.89 [-1.29, -0.50]	+1.55	0.07	×0.15
Post-hoc Reinhard [20]	-1.58 [-1.96, -1.19]	+1.53	0.12	×0.13

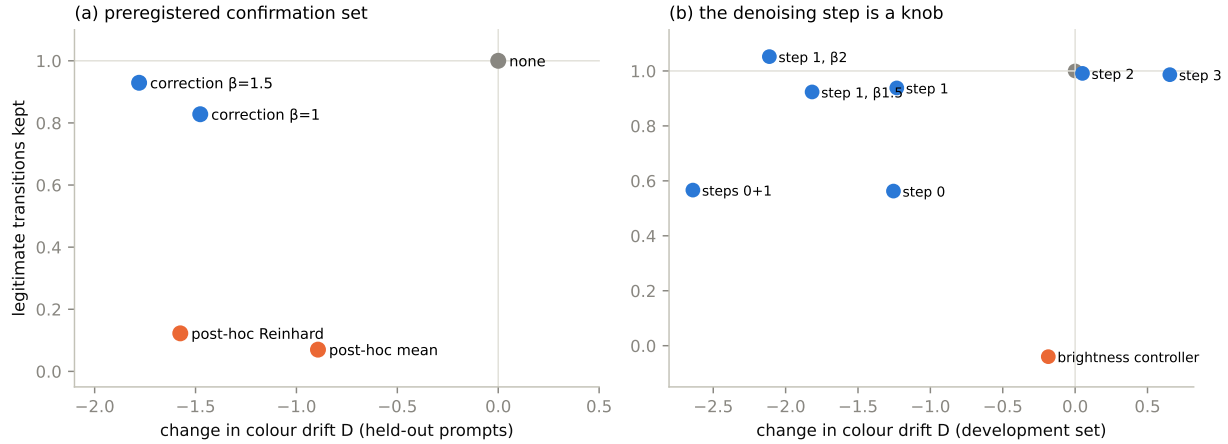


Figure 2: Drift reduction versus legitimate brightness transitions kept. (a) Preregistered confirmation set: only the content-aware correction is in the upper left. (b) Development set: the denoising step at which the anchor's tone is transferred trades drift removal against content fidelity.

This resolves an earlier puzzle: the sink's direct effect on the next chunk is tiny, yet it roughly halves drift. [Interventional test with sink 0 vs 3 from the relaxation study.]

6 Two levers

Raising α: a content-aware correction. At denoising step 1 of 4, the noisy chunk is run once more with a temporary KV cache that holds only the anchor. The difference between the per-channel spatial means (the DC tone) of that prediction and of the real one is added back, scaled by β. Nothing else changes, including the main cache and the random stream.

- **Confirmation (preregistered, 40 held-out prompts × 4 seeds).** Colour drift D falls by 1.48 $[1.19, 1.76]$ at $\beta = 1$ and by 1.78 at $\beta = 1.5$. Structure drift K is non-inferior. On held-out prompts with legitimate brightness changes, 83% / 93% of the change is kept.
- **Baselines.** Post-hoc colour matching to the anchor (mean, or Reinhard mean and standard deviation) reaches similar D (-0.89, -1.58) but keeps only 7% / 12% of legitimate transitions (Table 3).
- **The step is a knob.** Correcting at step 0 removes more drift but erases tunnels (56% kept); steps 2–3 do nothing.
- **Cost and transfer.** The correction costs one forward pass per chunk and some flicker. On LongLive it reduces D by half as much.

Moving μ: calibrated prompts.

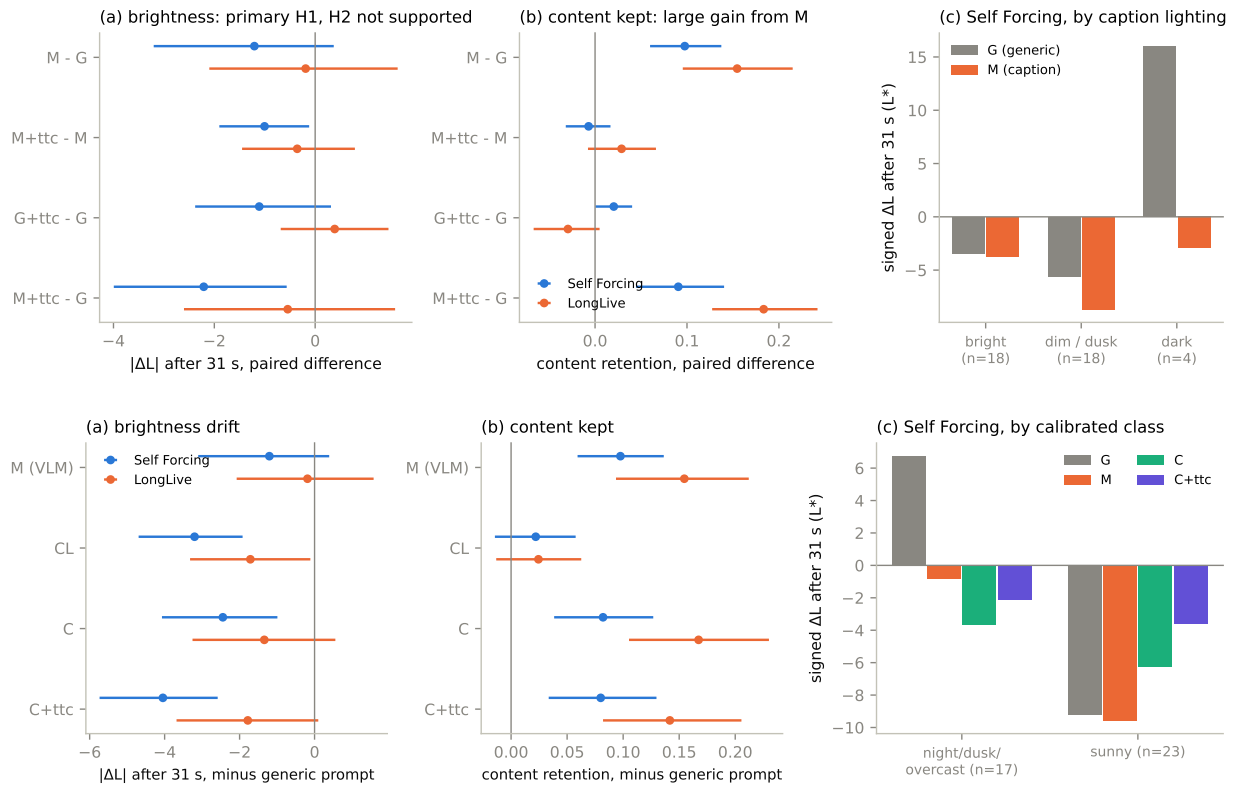


Figure 3: Moving μ. Top: free-form VLM captions (preregistered; not supported for brightness, large content gain). Bottom: brightness-calibrated lighting descriptions (preregistered; supported against the generic prompt on Self Forcing). Paired differences with opening-cluster bootstrap 95% CIs.

- **Free-form caption (preregistered).** Describing the opening with a Qwen2.5-VL [18] caption does not reduce brightness drift on either model (Self Forcing -1.21 [-3.09, 0.36]). The caption describes half of the normally lit frames as "dim" and pulls them darker. It does improve content retention ($+0.10$ / $+0.15$).
- **Calibrated lighting (preregistered).** Choosing the lighting sentence whose measured default is nearest the opening's brightness reduces $|\Delta L|$ by 26% on Self Forcing (Holm $p = 0.013$); with the correction, by 44%.
- **Calibration is per model.** On LongLive, the preregistered test was not significant even with LongLive's own calibration; the lighting-only prompt reduces drift by 34% (secondary).

(Fig. 3). [Out-of-sample prediction of these outcomes from α and μ.]

7 Context strategies

A preregistered factorial study (10 strategies, 40 prompts $\times$ 4 seeds) found:

- a 3-frame sink cuts colour drift by 7.56 [6.81, 8.32];
- window size matters only without a sink;
- RoPE re-alignment, which Self Forcing was not trained for, does not help;
- MUSIQ varies only between 66 and 69 across strategies whose drift differs threefold;
- the GPU architecture changes outputs, so 70% of nominally paired comparisons were not paired.

8 What did not hold

We report claims that were revised or withdrawn after checks:

- that MUSIQ rewards collapsed videos (it is merely insensitive);
- that the sink stabilises the default (trend difference +0.5 [-2.4, 3.5]);
- that the model follows its context by 75% (only for artificial shifts);
- that early dynamics forecast the final brightness;
- that describing the opening suffices.

9 Limitations

- All models but SkyReels share the Wan2.1-1.3B base.
- Openings are held images, not real video.
- The correction acts on colour only.
- Lighting calibration uses four discrete levels and cannot match openings brighter than the sunny default.

A Preregistrations, deviations and pitfalls

Table 4: The six preregistered studies. Code was frozen by SHA-256 hash before generation; studies 3–6 were also committed to the public repository before generation.

#	Study	Primary hypotheses	Outcome
1	Context strategies (2,000 videos)	sink lowers drift; window matters only without sink	supported
2	Content-aware correction (480 + 96 videos)	lower colour drift; structure non-inferior; transitions kept	supported
3	Caption prompts (800 videos)	caption < generic; +correction < caption	not supported
4	Calibrated lighting (560 videos)	calibrated < caption; calibrated < generic	supported vs generic (SF)
5	Calibrated lighting, LongLive's own levels	same, LongLive	not supported
6	Relaxation (interventional)	sink raises α (two models); correction raises α; sink raises α_b	[]

Deviations.

- In study 3, a preempted job resumed on another GPU type and split four LongLive openings across architectures. We regenerated the affected arms on the original architecture; the conclusions are unchanged (sensitivity analysis in the repository).
- In study 6, the ODE-initialised student was replaced by SiD v2 before generation, because it collapses to blur after the opening.
- Exploratory configurations (sink 1; correction on LongLive and CausVid) were added after freezing and before any data, and are reported as exploratory.

Pitfalls for evaluating video models.

1. **The GPU architecture changes the video.** With bf16, the same seed and code give bit-identical videos within an architecture but different videos across A40, A100/A800 and L40S/RTX 6000 Ada. Paired designs must pin the architecture.
2. **Resumed jobs can silently break pairing.** Generators that skip existing outputs resume a preempted job wherever it lands.
3. **Popular quality metrics miss drift.** MUSIQ, used by VBench for imaging quality, barely moves across strategies that differ threefold in colour drift.
4. **Natural variation is too small to estimate persistence.** Within a prompt, seeds start only about 2 L^* apart, so within-prompt slopes of end on start brightness have confidence intervals of ± 0.4. Controlled shifts of ± 16 L^* are needed.
5. **Change-on-baseline regressions can mislead.** Regressing change on an opening value measured with error produces spurious regression to the mean (Oldham's problem). Held openings are measured almost without error, and the interventional design removes the issue.

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